

Jeffrey Resto

Intrusion Detection System Configuration with Snort

Overview:

For this project, I configured and tested an Intrusion Detection System (IDS) within a Linux environment using Snort. The goal was to understand how Snort works, including how it monitors network traffic, applies detection rules, and reacts to attacks. I also wanted to see how encrypted traffic, such as HTTPS, affects Snort's visibility and how specific rules can be written to detect either keywords or behaviors. I also did a few network scans to observe how Snort logs and responds to suspicious activity in real time.

Tools & Environment:

- Labtainers (virtual lab)
- Snort
- Nmap for network scanning
- Linux operating system

Technical Tasks:

- Installed and configured Snort within Labtainers setup to act as an IDS.
- Created and tested custom Snort rules to detect certain traffic patterns and keywords.
- Simulated network scans using Nmap to view Snort alerts and ensure the rules work.
- Observed how secure protocols (HTTPS) limited Snort's ability.
- Used iptables to mirror network traffic for analysis and packet capture.
- Reviewed Snort logs to understand alert generation and potential false positives.

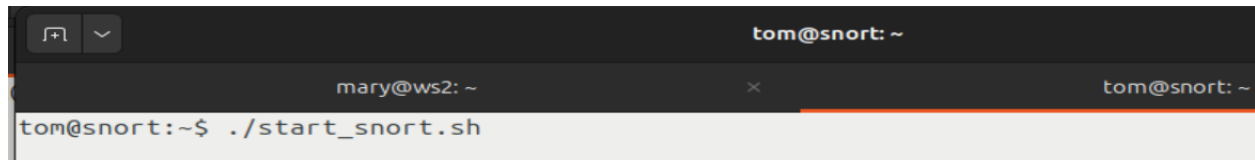
1. Launch Labtainer and Start the Snort Lab:

Starting Snort in Network Intrusion Detection Mode:

In the Snort component terminal, start Snort by executing:

```
./start_snort.sh
```

Snort will begin running and display alerts directly in the console.



```
tom@snort: ~  
mary@ws2: ~  
tom@snort:~$ ./start_snort.sh
```

2. Stopping Snort:

When you need to stop Snort (e.g., to modify rules), press CTRL+C.

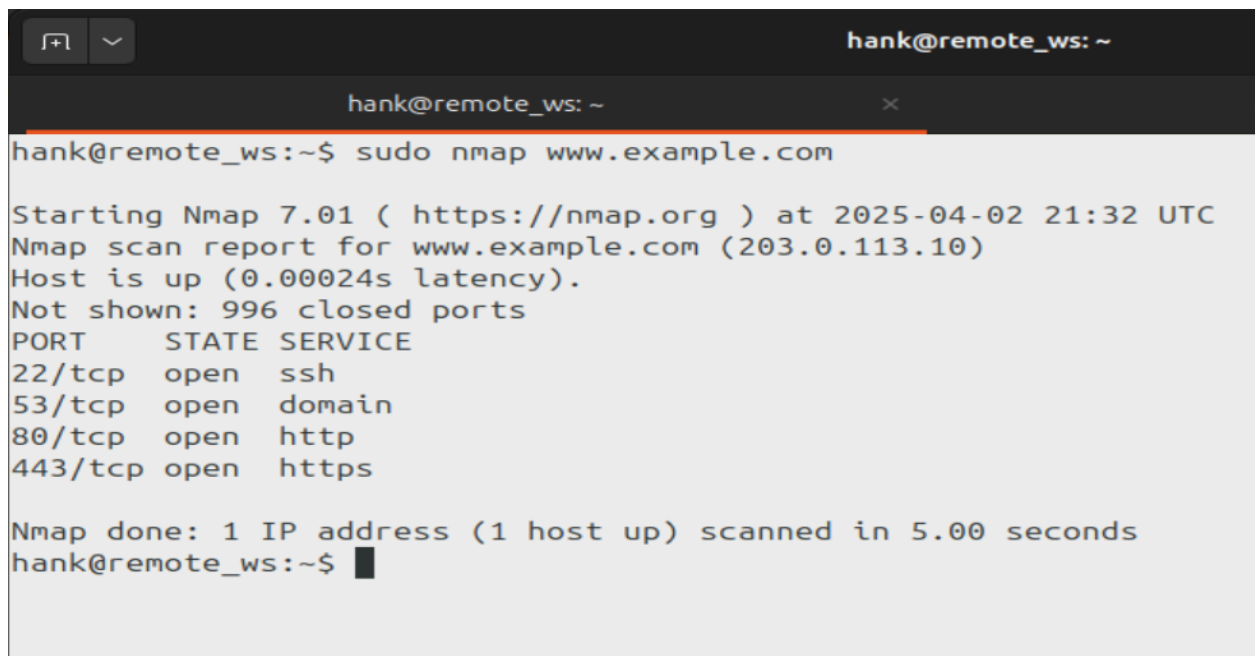
3. Understanding Pre-Configured Snort Rules

1. Run a Network Scan and Observe Alerts:

From the remote workstation, perform an nmap scan to trigger alerts:

```
sudo nmap www.example.com
```

Watch for alerts on the Snort console. Alerts generated by the preconfigured rules can be examined in the `/etc/snort/rules/` directory.



```
hank@remote_ws: ~  
hank@remote_ws:~$ sudo nmap www.example.com  
  
Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-02 21:32 UTC  
Nmap scan report for www.example.com (203.0.113.10)  
Host is up (0.00024s latency).  
Not shown: 996 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
53/tcp    open  domain  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 5.00 seconds  
hank@remote_ws:~$
```

```
tom@snort: ~  
mary@ws2: ~  
tom@snort:~$ ./start_snort.sh  
04/02-21:32:44.961675  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:44.961675  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:44.963222  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 203.0.113.20 -> 203.0.113.10  
04/02-21:32:47.874279  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64376 -> 203.0.113.10:705  
04/02-21:32:47.940581  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 203.0.113.20:64376 -> 203.0.113.10:161
```

4. Write and Test a Basic Custom Snort Rule

1. Create a Simple Rule to Detect All TCP Traffic:

Stop Snort by pressing CTRL+C.

Open the custom rule file:

```
sudo nano /etc/snort/rules/local.rules
```

Add the following rule to alert on every TCP packet:

```
alert tcp any any -> any any (msg:"TCP detected"; sid:00002;)
```

```
mary@ws2: ~  
tom@snort: ~  
GNU nano 2.5.3 File: /etc/snort/rules/local.rules  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
# -----  
# LOCAL RULES  
# -----  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert tcp any any -> any any (msg:"TCP detected";sid:00002;)
```

Save the file and restart Snort by running `./start_snort.sh`.

```
tom@snort:~$ ./start_snort.sh
```

2. Test the Rule:

Use Firefox on the remote workstation to visit www.example.com.



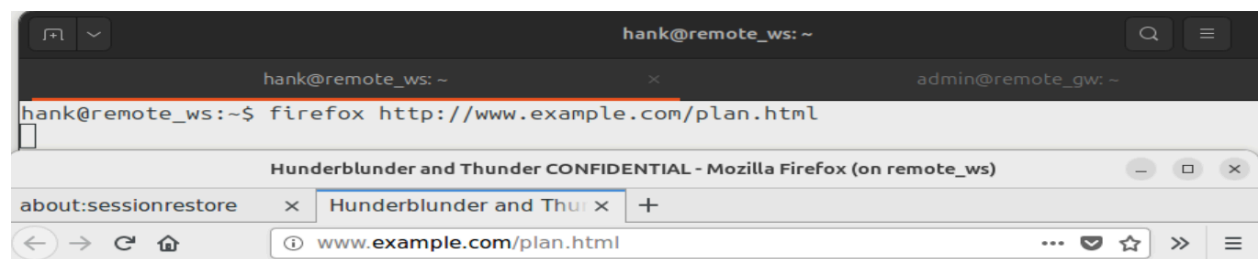
Observe the Snort console for numerous alerts. Once confirmed, stop Snort and delete this rule to avoid excessive alerts.

```
tom@snort:~$ ./start_snort.sh
04/02-21:42:54.367368  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:54.367678  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:54.368613  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:56.191878  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:56.192107  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:56.214354  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
04/02-21:42:56.214753  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:57.879102  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 203.0.113.20:49686 -> 203.0.113.10:80
04/02-21:42:57.882809  [**] [1:2:0] TCP detected [**] [Priority: 0] {TCP} 192.168.1.2:80 -> 192.168.1.10:49686
```

5. Create a Custom Rule to Detect "CONFIDENTIAL" Traffic

1. View the Confidential Page:

Use Firefox to open <http://www.example.com/plan.html> on the remote workstation to see a confidential document.



Hunderblunder and Thunder Turkey Ranch

Startup Business Plan: CONFIDENTIAL

First we get some turkey eggs and figure out how to hatch them. Then we get some turkey feed. Then we find an ax. OK, I did my bit, you guys run with it.

2. Add a Rule to Detect Confidential Content:

Open the custom rule file and add a rule to trigger an alert when

"CONFIDENTIAL" is detected:

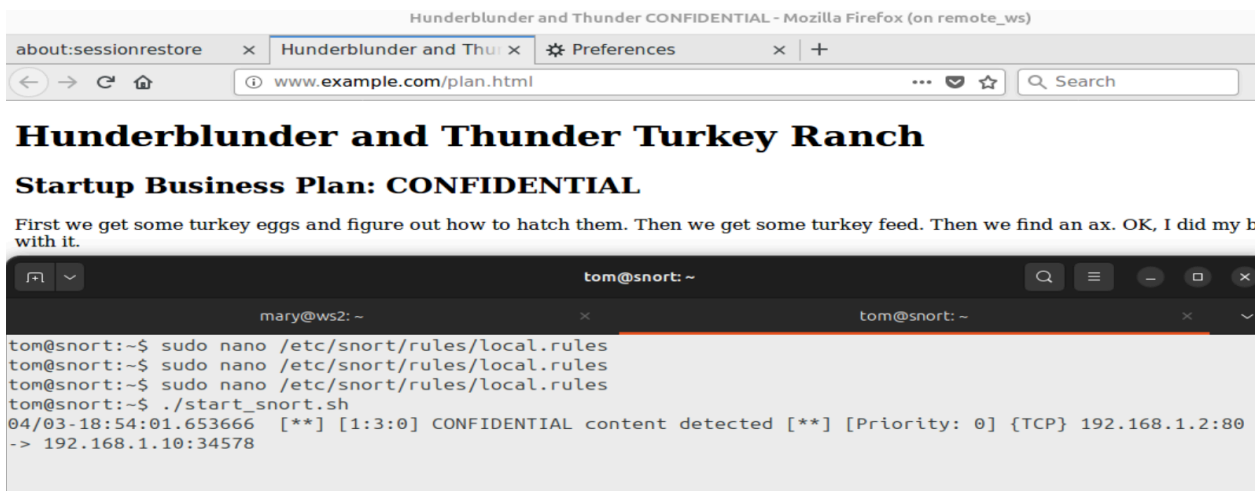
alert tcp any any -> any any (msg:"CONFIDENTIAL content detected";

content:"CONFIDENTIAL"; sid:00003;)

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any any (msg:"CONFIDENTIAL content detected"; content:"CONFIDENTIAL"; sid:00003;)
```

Restart Snort and clear the Firefox browsing history (Menu > Preferences > Security & Privacy).

Refresh plan.html in Firefox to trigger the alert.



The screenshot shows a Firefox browser window with the address bar at `www.example.com/plan.html`. The page content includes the heading **Hunderblunder and Thunder Turkey Ranch** and a section titled **Startup Business Plan: CONFIDENTIAL**. The text below this section reads: "First we get some turkey eggs and figure out how to hatch them. Then we get some turkey feed. Then we find an ax. OK, I did my l with it."

Below the browser window, a terminal window shows the following commands and output:

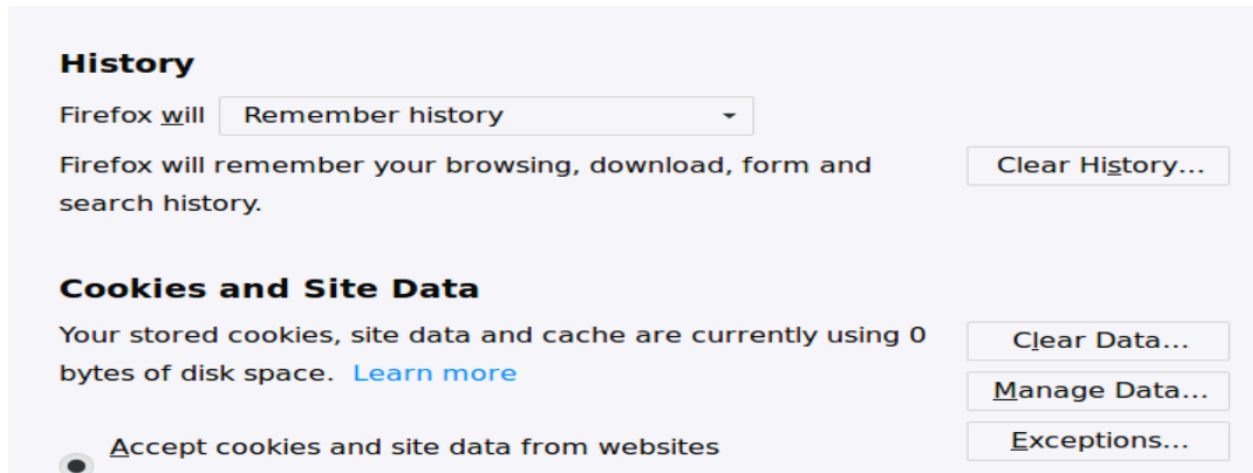
```
tom@snort:~$ sudo nano /etc/snort/rules/local.rules
tom@snort:~$ sudo nano /etc/snort/rules/local.rules
tom@snort:~$ sudo nano /etc/snort/rules/local.rules
tom@snort:~$ ./start_snort.sh
04/03-18:54:01.653666  [**] [1:3:0] CONFIDENTIAL content detected [**] [Priority: 0] {TCP} 192.168.1.2:80
-> 192.168.1.10:34578
```

6. Explore Effects of Encryption on Intrusion Detection

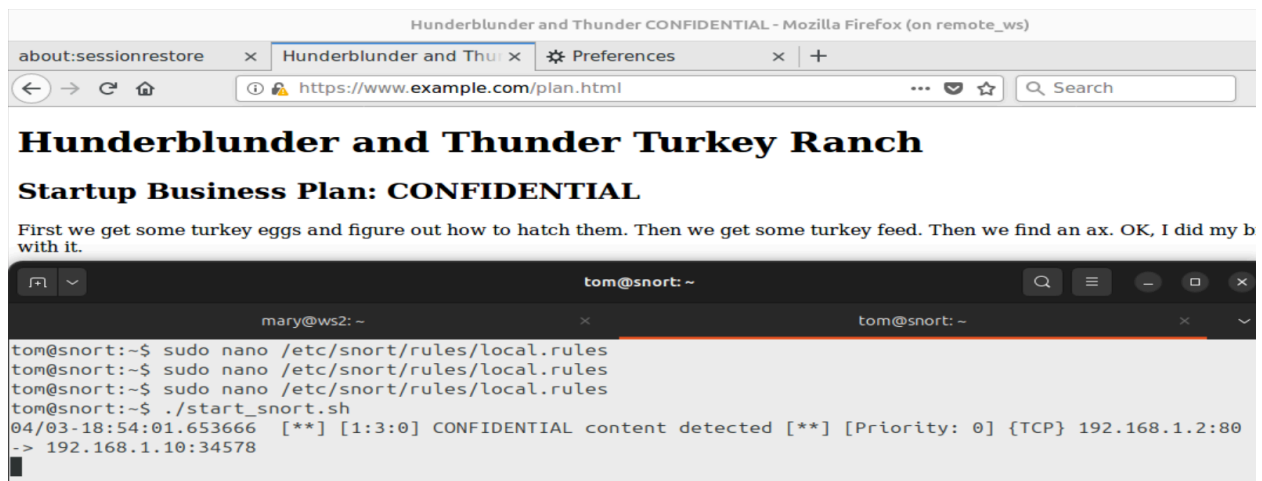
1. Access the Confidential Page via HTTPS:

Clear the browsing history again, then navigate to

<https://www.example.com/plan.html>.



Observe the Snort console. You should not see a new alert because HTTPS encrypts the content, making it unreadable by Snort.



7. Monitor Internal Traffic

1. Run nmap on an Internal Workstation (ws2):

Go to the ws2 terminal and execute:

`sudo nmap www.example.com`

```
mary@ws2: ~  
mary@ws2:~$ sudo nmap www.example.com  
  
Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 19:44 UTC  
Nmap scan report for www.example.com (192.168.1.2)  
Host is up (0.00034s latency).  
Not shown: 996 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
443/tcp   open  https  
3306/tcp  open  mysql  
  
Nmap done: 1 IP address (1 host up) scanned in 10.39 seconds  
mary@ws2:~$
```

No alert should appear on the Snort console because internal traffic is not yet mirrored.

```
mary@ws2: ~ tom@snort: ~  
tom@snort:~$ ./start_snort.sh  
04/03-19:44:08.902098  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority  
: 3] {ICMP} 192.168.1.2 -> 192.168.2.1  
04/03-19:44:08.903702  [**] [1:451:5] ICMP Timestamp Reply [**] [Classification: Misc activity] [Pri  
ority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1
```

2. Mirror Internal Traffic to Snort:

On the gateway component, edit the `/etc/rc.local` file to add the following line:

```
iptables -t mangle -A PREROUTING -i $lan2 -j TEE --gateway
```

192.168.3.1

Apply the new settings:

```
sudo /etc/rc.local
```

```
admin@web_server:~  
ubuntu@gateway: ~  
GNU nano 2.5.3 File: /etc/rc.local  
#  
iptables --flush  
iptables -t nat --flush  
iptables -t mangle --flush  
iptables --delete-chain  
iptables -t nat --delete-chain  
iptables -t mangle --delete-chain  
#  
# mirror incoming wan traffic to snort  
#  
iptables -t mangle -A PREROUTING -i $wan -j TEE --gateway 192.168.3.1  
iptables -t mangle -A PREROUTING -i $lan1 -j TEE --gateway 192.168.3.1  
iptables -t mangle -A PREROUTING -i $lan2 -j TEE --gateway 192.168.3.1  
  
ubuntu@gateway:~$ sudo nano /etc/rc.local  
ubuntu@gateway:~$ sudo /etc/rc.local  
[ ok ] Restarting dnsmasq (via systemctl): dnsmasq.service.  
SIODELRT: No such process  
ubuntu@gateway:~$
```

Restart Snort, then rerun the nmap scan from ws2 to see alerts generated for internal traffic.

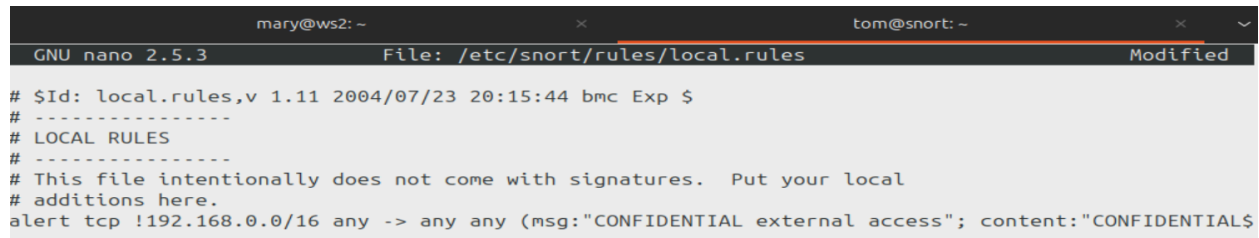
```
mary@ws2:~$ sudo nmap www.example.com  
  
Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 20:02 UTC  
Nmap scan report for www.example.com (192.168.1.2)  
Host is up (0.00036s latency).  
Not shown: 996 closed ports  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
443/tcp   open  https  
3306/tcp  open  mysql  
  
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds  
  
tom@snort:~$ ./start_snort.sh  
04/03-20:02:04.391482  [**] [1:469:3] ICMP PING NMAP [**] [Classification: Attempted Information Leak] [Priority: 2] {ICMP} 192.168.2.1 -> 192.168.1.2  
04/03-20:02:04.391482  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2  
04/03-20:02:04.391603  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1  
04/03-20:02:04.392825  [**] [1:453:5] ICMP Timestamp Request [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.2.1 -> 192.168.1.2  
04/03-20:02:04.392910  [**] [1:451:5] ICMP Timestamp Reply [**] [Classification: Misc activity] [Priority: 3] {ICMP} 192.168.1.2 -> 192.168.2.1  
04/03-20:02:07.380308  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:60102 -> 192.168.1.2:161  
04/03-20:02:11.129612  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 192.168.2.1:60102 -> 192.168.1.2:705
```

8. Configure Snort to Differentiate Between Internal and External Access

1. Update Snort Rules for Internal and External Access:

Modify the rule in local.rules to trigger alerts only when external addresses access the "CONFIDENTIAL" content:

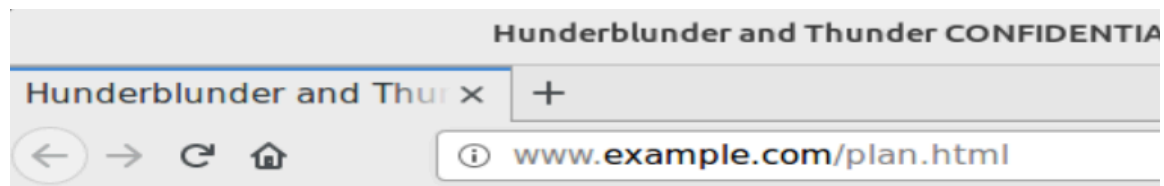

```
alert tcp !192.168.0.0/16 any -> any any (msg:"CONFIDENTIAL external access";  
content:"CONFIDENTIAL"; sid:00004;)
```



```
mary@ws2: ~  
tom@snort: ~  
GNU nano 2.5.3 File: /etc/snort/rules/local.rules Modified  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
# -----  
# LOCAL RULES  
# -----  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert tcp !192.168.0.0/16 any -> any any (msg:"CONFIDENTIAL external access"; content:"CONFIDENTIAL$
```

Restart Snort and test each scenario:

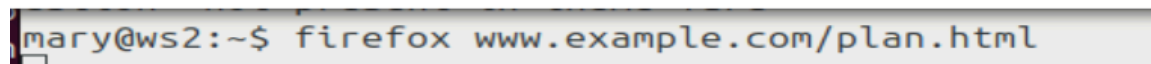
External access to plan.html should trigger an alert.



Hunderblunder and Thunder

Startup Business Plan: CONFIDENTIAL

First we get some turkey eggs and figure out how to hatch them
an ax. OK, I did my bit, you guys run with it.



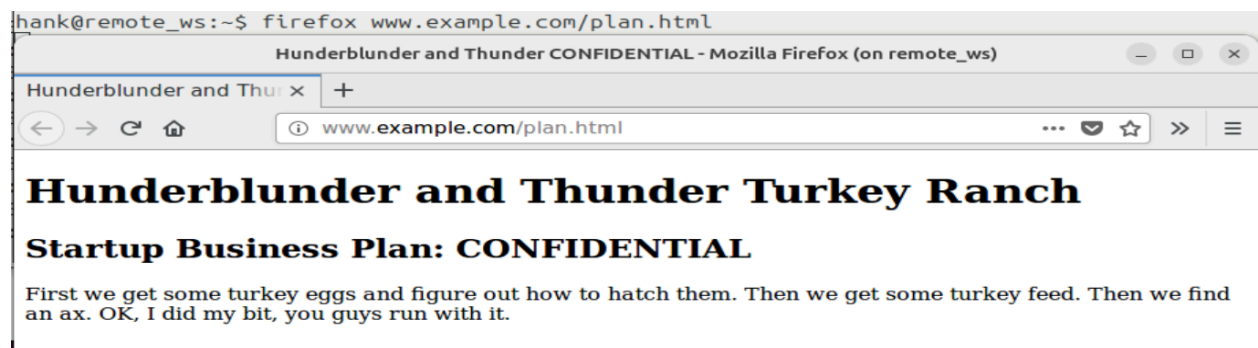
```
mary@ws2:~$ firefox www.example.com/plan.html
```

```

04/03-20:11:40.303867  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:41.341235  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:42.107775  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.419290  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.447513  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:45.561267  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:46.344821  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:47.112621  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.421862  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.453025  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:50.562962  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:11:51.345660  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifi
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10

```

Internal access to plan.html should not.



```

tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:35.606734  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:37.101370  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10
04/03-20:12:42.107889  [**] [1:402:7] ICMP Destination Unreachable Port Unreachable [**] [Classifica
tion: Misc activity] [Priority: 3] {ICMP} 203.0.113.1 -> 203.0.113.10

```

Both internal and external nmap scans should generate alerts.

```

hank@remote_ws:~$ sudo nmap www.example.com

Starting Nmap 7.01 ( https://nmap.org ) at 2025-04-03 20:15 UTC
Nmap scan report for www.example.com (203.0.113.10)
Host is up (0.00075s latency).
Not shown: 996 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 9.51 seconds

```

```
04/03-20:12:25.598882 04/03-20:12:27.090963 04/03-20:12:30.602653 04/03-20:12:32.096579 04/03-20:12:35.606734 04/03-20:12:37.101370 04/03-20:12:42.107889 04/03-20:15:42.361959 04/03-20:15:42.361959 04/03-20:15:42.363473 04/03-20:15:43.698822 04/03-20:15:45.153375
```

stoplab snort